

Capability Emergence Can Be Forecast: Per-Seed, In Advance, With Calibrated Intervals, Certified False Alarms, and a Blind Pre-Registered Gate

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Abstract

Emergent capabilities are widely treated as unpredictable: aggregate loss improves smoothly while specific abilities appear abruptly. Prior work offers early-warning *indicators* but never scores them as *forecasts* — no lead time at a controlled false-alarm rate, no calibration, no negative controls, no blind validation. We supply that discipline and show that, in controlled settings spanning grokking model systems and small language models, emergence *timing* is forecastable per training run, in advance, with calibrated uncertainty. On language models: across 30 transformers at identical configuration — where the strongest published baseline predicts a single constant [1] — the formation time of the previous-token head (the mechanistic precursor of induction [2]) forecasts each seed’s induction emergence at Spearman $\rho = 0.977$ (95% CI [0.911, 0.995]) with median lead 975 steps ($\sim 15\%$ of training). A best-case training-loss rule *ties* that ranking ($\rho = 0.977$) but with 50-step median lead — a nowcast, not a forecast. Split-conformal intervals of width 300 steps, anchored at the precursor alarm, covered 15/15 held-out seeds at 90% nominal. The entire rule was then frozen verbatim — alarm threshold, offset 975, quantile 150, commit-stamped — and subjected to a blind pre-registered gate on a never-seen configuration (wider model, sparser repetition): it covered 10/10 runs (bar 7/10), alarmed pre-event 10/10, and false-alarmed on 0/3 fresh negatives; the precursor-to-emergence gap was approximately configuration-invariant (975 $\rightarrow$ 1,012 steps). All false-alarm claims are certified against 15+3 *manufactured capability-blocked negatives* (data-ablated and architectural) — a resource we argue is mandatory for capability monitoring and that no public suite provides. Public-checkpoint evidence (Pythia): the precursor leads the induction cliff by a full checkpoint stage in all 5 suites probed, and all five cliff at the same token count, confirming that public suites cannot test per-seed forecasting. The grokking half of the paper (466 runs) contributes the evaluation methodology and its hard lessons: a robustness-versus-false-alarm tradeoff (the context features that suppress false alarms are exactly what breaks under intervention), mechanism-factored gates that resolve it, non-transfer of fitted calibration across regimes, and a label-free probe family that died under pre-registered confirmation. Three kill criteria fired and are reported. Every prediction, freeze, and verdict in both halves is commit-stamped before its data existed. Scope: named capabilities with known mechanistic precursors, small models; we state what stands between this and frontier deployment.

1 Introduction

When a capability emerges during training, the training curve rarely announces it in advance: loss declines smoothly while the ability arrives abruptly [2, 3]. For frontier-scale systems this unpredictability is a safety problem with a name — labs want to know what a model will be able to do *before* it can do it. A literature of early-warning signals has grown in response: loss-curve

oscillations that predict whether grokking will occur [4], spectral observables that show precursors of phase transitions [5], geometric stage detection [6], and scale-axis extrapolations [7, 8]. What none of this work does is score a *forecast*: state, in advance, when a capability will arrive; attach calibrated uncertainty; measure lead time at a controlled false-alarm rate against negatives where the capability never arrives; and validate the frozen system blind. Indicators without that discipline are weather lore, not weather forecasting.

This paper does two things. **First**, it builds the missing evaluation discipline on a grokking model system — 466 training runs across two domains with interventions that shift emergence timing $\sim 100\times$ — and reports what the discipline kills: single-signal alarms die on look-alike negatives; the context features that control false alarms are exactly the features that break under intentional shift (a robustness–false-alarm tradeoff); fitted calibration does not transfer across regimes; and an initially spectacular label-free probe family failed its pre-registered confirmation. **Second**, it carries the surviving methodology to language models and obtains the result the discipline was built for (Figure 1): per-seed forecasts of induction-head emergence from a mechanistic precursor, with calibrated intervals, certified false alarms, and a blind pre-registered transfer gate passed at ceiling.

Contributions, each with its adversarial control:

1. **A forecasting benchmark and protocol for emergence timing**: event definitions robust to outliers, lead time at capped false-alarm rate, champion selection on training data only, (correlation, lead) reported as a pair — plus a *negative taxonomy* (budget-censored vs. structurally blocked vs. *manufactured*) and a minimum-negative-count rule learned the hard way.
2. **The per-seed result** (Sections 5.2–5.3): at fixed configuration, where the config-equation baseline of Aoyama and Wilcox [1] is a constant by construction, precursor formation time forecasts emergence at $\rho = 0.977$ with 975-step median lead; a best-case loss rule ties the correlation with 50 steps of lead. Warning time lives in *where you anchor*, and only the circuit precursor anchors early.
3. **Manufactured negatives** (Section 5.2): capability-blocked training runs — data-ablated (repetition stripped) and architectural (one layer) — enabling the first certified false-alarm rates for emergence forecasting (0/10 and 0/10 on data-ablated, 0/5 on architectural, 0/3 blind).
4. **The blind gate** (Section 5.4): the full rule frozen verbatim and commit-stamped, then tested on a never-seen configuration: 10/10 interval coverage, 10/10 pre-event alarms, $\rho = 0.988$. The precursor-to-cliff gap is approximately config-invariant across the shift we tested — an unexplained regularity we flag as an open question.
5. **The kill ledger** (Section 6): three pre-registered kill criteria fired across the program and are reported with their mechanisms, alongside every failed prediction. The positive results above survived the same regime that produced these deaths; that, not any single number, is the paper’s claim to trust.

Everything is pre-registered in a public commit chain (every freeze precedes its data; Section 9), and the corpus, fleets, and harness are released as a benchmark.

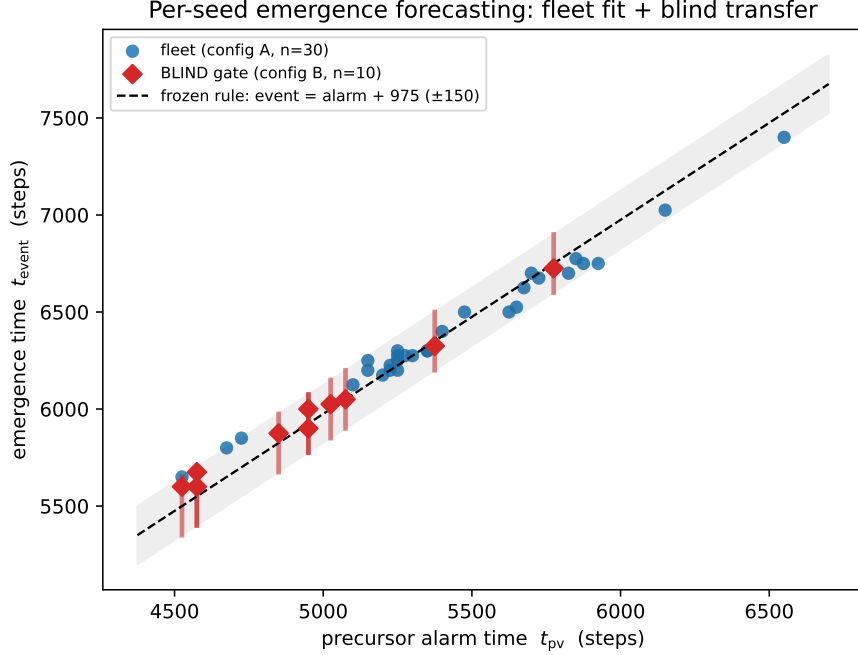


Figure 1: **The central result.** Precursor alarm time vs. emergence time. Blue: the calibration fleet (30 seeds, config A) — Spearman $\rho = 0.977$. The dashed line and band are the *frozen* rule (event = alarm + 975 $\pm$ 150). Red diamonds with bars: the blind gate — 10 fresh seeds on a never-seen configuration (wider model, sparser repetition), scored once against the frozen rule: 10/10 covered. Negatives (18 manufactured capability-blocked runs, not shown) produced zero false alarms.

2 Related work

Notsawo et al. [4] predict *whether* grokking will occur from early loss-curve oscillations — a scalar whether-classifier; its oscillation feature is one of our baselines. Anonymous et al. [5] derive spectral early-warning indicators (2-RDM heat capacity) across four transition settings but report no lead time, calibration, false-alarm rate, or baseline race; our label-free rung (Section 6) shows concretely why unscored spectral indicators can mislead: ours died under confirmation. Hoogland et al. [6] detect developmental stages retrospectively via the local learning coefficient. Aoyama and Wilcox [1] predict induction-head formation time from pre-training config (batch, context, bigram statistics) — powerful, but constant per config; our fleet result answers precisely the variance their equation cannot (per-seed timing at fixed config), and our Pythia analysis confirms their account across public suites (all five cliff at the same token count). Scale-axis emergence prediction [7, 8] forecasts across model size, orthogonal to our within-run time axis. Olsson et al. [2] established the induction-head phase change and its previous-token precursor; Barak et al. [9] and progress measures [10] established hidden progress; we turn these observations into scored forecasts. The grokking substrate builds on [11–13] and our companion papers [14–16], which established the representational-timing mechanism and the norm clock that our two-gate forecaster exploits.

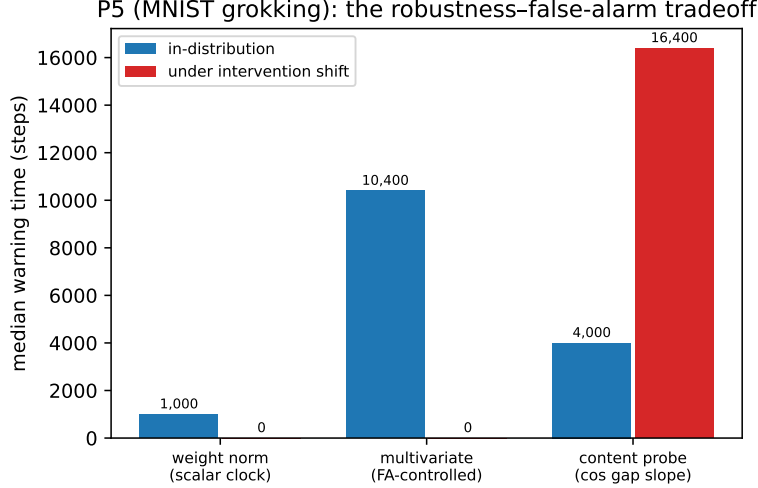


Figure 2: **The robustness-false-alarm tradeoff** (MNIST grokking; median warning time at $\leq 5\%$ FA, higher is better). The scalar clock (weight norm) forecasts in-distribution but collapses when interventions decouple the norm from the outcome. The multivariate forecaster that best controls false alarms in-distribution (10,400 steps, 60% of the delay) collapses hardest under shift — its norm-context features are exactly what the intervention breaks. The bare content probe survives the shift (16,400 steps) but is FA-fragile in corpora with look-alike negatives.

3 Forecasting emergence: definitions and discipline

A **run** logs behavioral and internal signals at fixed cadence. An **event** is the first crossing of an absolute behavioral criterion robust to outliers (never a fraction of a run’s own maximum — a rule adopted after that operationalization failed twice; Section 6). A **forecaster** observes the log prefix and may **alarm** once; its **lead** is $t_{\text{event}} - t_{\text{alarm}}$ (alarms after the event score zero; no alarm on an eventing run is a miss). **False-alarm (FA) rate** is the fraction of *negative* runs — runs where the capability never arrives — that alarm. Thresholds and champions are selected on training runs only, under an FA cap ($\leq 5\%$), with at least five negatives on every side of every split (a minimum learned from a degenerate fit, Section 6). Negatives come in three kinds: *budget-censored* (would event with more budget — these legitimately attract alarms), *structurally blocked* (wrong learned structure), and *manufactured* (constructed so the capability cannot form). Forecast quality is always a *pair* — (ranking skill, lead) — because a rule can rank perfectly by detecting the event as it happens; and point forecasts carry split-conformal intervals whose empirical coverage is reported. Blind validation means: every constant frozen and commit-stamped, then fresh runs, one scoring pass.

4 Part I: the grokking benchmark, and what the discipline kills

We built the protocol on 466 pre-existing grokking runs (modular arithmetic transformers; MNIST MLPs in the Omnigrok regime) whose companion-paper interventions (contrastive priors, norm clamps) shift emergence timing up to $\sim 100\times$ and — crucially — *decouple* the dominant scalar clock (weight norm) from the outcome. Full protocol and verdicts are in the repository (`plan_p5.md`, `P5_RESULTS.md`); four results shape everything downstream.

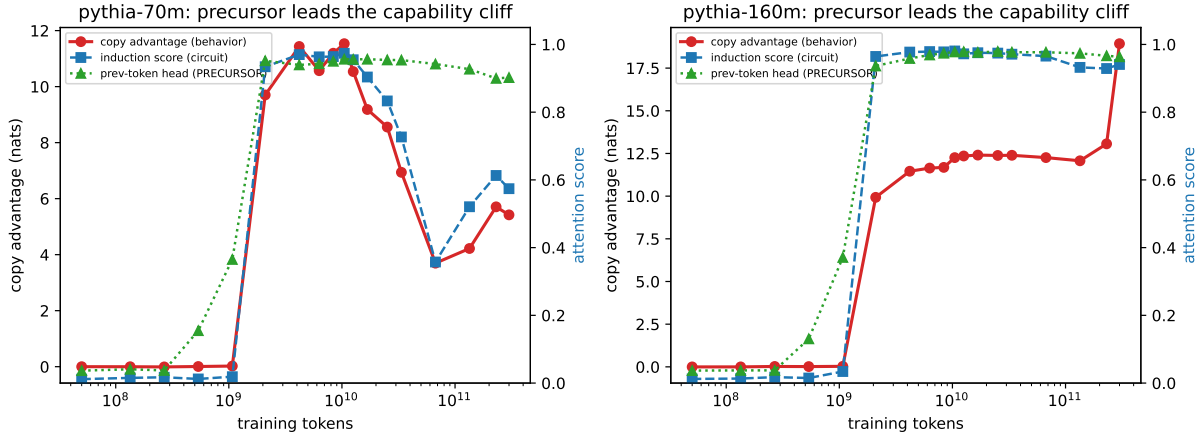


Figure 3: **Public Pythia checkpoints (models we did not train).** The previous-token head (green) rises a full public-checkpoint stage before the induction-head score (blue) and the behavioral copy advantage (red) cliff together. Left: pythia-70m; right: pythia-160m. The 70m model also partially *loses* induction late in training (prefix 0.93→0.36 at 67B tokens) — capability regression exists and is monitorable.

(i) **In-regime forecasting works.** A multivariate forecaster with mechanism-motivated features achieved 10,400-step median lead (60% of the delay) on held-out seeds at zero structural false alarms, and 18,000 steps prospectively on pre-registered fresh runs including a never-seen norm pin (bar 5,200) — all four prospective predictions passed.

(ii) **The robustness–false-alarm tradeoff** (Figure 2). Trained on control runs only and tested on intervention arms, the scalar norm clock collapses (median lead 0; the interventions hold the norm where controls never generalize), and the FA-controlled multivariate collapses with it — *because* the context features that suppress false alarms are the features the intervention breaks. The bare content probe survives (4,400–16,400 steps). A monitor tuned quiet on its training distribution goes blind exactly when the recipe changes.

(iii) **Mechanism-factored gates resolve the tradeoff where mechanism is known.** Composing a content gate with an *a-priori* norm-viability window (constants from the companion papers, not fit) retains 12,400 steps (53% of the delay) under the same shift at zero miss. The portable ingredient is mechanism, not fitted calibration: fitted thresholds transferred across domains in neither direction (kill K5).

(iv) **The discipline kills seductive numbers.** An algorithmic-domain corpus with structure-complete-but-blocked negatives admitted *no* forecaster at 5% FA (kill K1/K8): trap negatives carry more “progress” signal than true positives show before their events. And a label-free spectral probe that posted a 34,200-step validated lead died under pre-registered confirmation — it false-alarmed on fresh structural negatives and, per-run, was largely detecting the *intervention*, not the emergence (kill K9). Both deaths are in the ledger (Section 6) and both shaped the LM protocol.

5 Part II: language models

5.1 Public checkpoints: the precursor leads, and why public suites are not enough

We probed 5 Pythia suites [17] (70m/160m/410m, standard and deduped) across training checkpoints for three quantities: behavioral copy advantage on repeated random sequences, the maximum per-head prefix-matching (induction) score, and the maximum early-layer previous-token score — the known mechanistic precursor [2]. The phase change is crisp in every suite (Figure 3): at 1.07B tokens copy advantage is 0.022 and the precursor is already at 0.37 ($\sim 10\times$ baseline) while induction sits at 0.019; one stage later both cliff (copy 9.7 nats, induction 0.93). Against a best-case loss threshold granted an oracle sweep, the precursor alarmed strictly earlier on 4/5 suites and never post-event (leads 1–8 stages); our pre-registered prediction here failed in both directions — the precursor led by *more* than predicted and loss was *stronger* than predicted (one stage of genuine timing) — and is reported as registered. Two structural facts matter more than the scores. First, all five suites, which share batch and context configuration, cliff at the same ~ 2.1 B tokens — consistent with config-determined timing [1] and fatal to using public suites for per-seed forecasting: there is one run per config. Second, an event definition we had frozen (50% of each run’s maximum) was silently broken by a late-training outlier in one suite — the same operationalization failure our grokking rungs had already punished — and was replaced, in print, by absolute criteria everywhere.

5.2 The seed fleet: per-seed forecasts, certified against manufactured negatives

Public suites cannot answer the forecasting question; fleets can. We trained 45 two-layer transformers on a fixed synthetic language (bigram-structured, with variable-offset within-context repetition so that positional shortcuts cannot substitute for induction — a shortcut our first design permitted, caught in smoke testing): 30 positives at one configuration (seeds vary; config fixed), plus 15 **manufactured negatives** — 10 *data-ablated* (identical language, zero repetition: induction has no training signal) and 5 *architectural* (one layer: induction is structurally impossible [2]). Events use an absolute criterion (copy advantage ≥ 2 nats, two consecutive evals).

Results, scored once against the pre-registered spec. Per-seed event times span 5,650–7,400 steps (spread $1.31\times$; the pre-registered bar of $1.3\times$ was cleared by 0.01 — variance at fixed config is real but modest, and we do not dramatize it). The precursor’s crossing time forecasts each seed’s event: $\rho = 0.977$ (bootstrap CI [0.911, 0.995], $n = 30$), median lead 975 steps (825–1,125) — $\sim 15\%$ of the run issued as advance notice, seed by seed, where the config equation predicts one constant. The best-case loss threshold ($\theta = 2.077$, granted an oracle sweep) *ties* the ranking ($\rho = 0.977$) — and forecasts nothing: its median lead is 50 steps (minimum -25), i.e., it detects the cliff in progress. Our pre-registered comparison metric (rank correlation alone) was therefore *wrong as designed*; the two-dimensional truth — equal correlation, $20\times$ the lead — is reported alongside the registered failure, and (correlation, lead) pairs are now the protocol’s required form. A second dissociation: the graded *behavioral* ramp that precedes the cliff carries no timing information ($\rho = -0.03$) — circuits time the transition; early behavior does not.

The negatives certified the false-alarm side. On data-ablated runs the precursor never formed (max layer-0 previous-token score 0.036–0.061 vs. threshold 0.10): bare precursor FA 0/10, conjunction FA 0/10, with pre-event alarms on 30/30 positives. Here our pre-registered mechanistic prediction was wrong and is reported: we expected the precursor to form anyway (predicting false alarms that would require a conjunctive gate); in fact a pure bigram language is solvable through the embedding pathway alone, so previous-token attention only pays where repetition exists. The manufactured negatives thus *certified* the precursor rather than trapping it — certification is the

other thing negative sets are for — and constructing a language where the trap is real (where previous-token context pays independently of repetition) is the named follow-up. Architectural negatives: zero events, induction score ≤ 0.014 throughout, zero alarms.

5.3 Calibrated intervals

Single thresholds are fragile (a P5 lesson bought with a dead result); forecasts should be intervals with measured coverage. Split-conformal calibration (seeds 1–15) around the anchored point forecast $t_{pv} + 975$ yields intervals of median width 300 steps that covered 15/15 held-out seeds (nominal 90%) — issued 975 steps before the event. The identical machinery anchored at the loss crossing covers too — with 50 steps of lead. Anchoring, not calibration, is where warning time comes from.

5.4 The blind gate

Everything above is one configuration. The ship-gate: freeze the entire rule verbatim — alarm at layer-0 previous-token score ≥ 0.10 ; forecast the event in $[t_{alarm} + 825, t_{alarm} + 1125]$ — commit it, then train 10 fresh seeds and 3 fresh data-ablated negatives on a configuration the rule never saw (model width $256 \rightarrow 320$; repetition density $0.75 \rightarrow 0.6$), and score once. Pre-registered expectations were deliberately humble: the P5 transfer kill (fitted calibration did not survive regime shift) made the frozen offset the component we flagged as most at risk. Outcome (Figure 1): 10/10 events; 10/10 pre-event alarms; **10/10 interval coverage** against a bar of 7/10; $\rho = 0.988$ within the new config; 0/3 false alarms; median lead 1,012 steps. The precursor-to-emergence gap barely moved under the shift ($975 \rightarrow 1,012$ steps) — our flagged risk did not materialize, and we do not fully understand why: whether the gap is set by optimization dynamics rather than configuration is, we believe, a sharp open question this benchmark makes askable.

6 The kill ledger

Positive results are only as credible as the regime that tried to kill them. Everything below fired as pre-registered and is reported with its mechanism:

kill / failure	where	mechanism
K1/K8: no forecaster at 5% FA	grokking (algorithmic)	trap negatives out-signal true positives
K5: calibration non-transfer	grokking (cross-domain)	fitted thresholds are regime-local
K9: label-free probe dies	grokking (confirmation)	signal was the intervention, not emergence
P1 (both clauses)	Pythia	precursor earlier, loss stronger than predicted
50%-of-max event rules	twice	fragile to outliers; absolute criteria adopted
P2c metric as registered	fleet	rank correlation without lead rewards nowcasts
P2d interior mechanism	fleet	bigram languages need no attention; negatives certified
single-negative FA cap	grokking (R5)	degenerate fit; minimum-negative rule adopted

Three of these (K1/K8, K5, K9) are full kill criteria; the rest are registered predictions that failed informatively. The protocol that survived this ledger is the one the language-model results were scored under.

7 Discussion: what this does and does not establish

Established. For a named capability with a known mechanistic precursor, in small transformers: emergence timing is forecastable per run, $\sim 15\%$ of training in advance, with calibrated intervals (15/15 and 10/10 coverage), certified false-alarm rates (0 across 18+3 manufactured negatives), a baseline accounting showing loss curves carry ranking but no lead, and blind transfer across a config shift. The evaluation discipline — manufactured negatives, (correlation, lead) pairs, FA caps with minimum negative counts, absolute event criteria, prereg-gated freezes — is domain-general and, we argue, the minimum standard any early-warning claim should meet.

Not established. Frontier applicability. The gap has named parts: (1) *precursor knowledge* — induction heads have a known antecedent; most capabilities of concern do not, and finding precursors is mechanistic-interpretability work our benchmark can score but not replace; (2) *negatives at scale* — frontier labs have $n=1$ runs and no capability-blocked counterfactuals; our manufactured-negative recipe (ablate the training signal the capability needs; block the architecture) is implementable in principle at any scale, and we regard a library of blocked runs as the single most actionable transfer of this work; (3) *recipe shift* — we passed one blind config shift on two axes; the P5 tradeoff says shift-robustness must be re-certified per monitor, and mechanism-factored designs are the only pattern we found that survives shift by construction; (4) *the gap-origin question* — the config-invariance of the precursor-to-cliff gap is unexplained; if it is an optimization-dynamics constant, calibration may transfer far more broadly than P5’s kill suggests, and if not, per-config calibration fleets are the cost of coverage. Emergence forecasting in the wild also faces capabilities that *regress* (pythia-70m’s late induction loss) and metric-induced discontinuities [3]; our absolute-criterion and interval machinery handles both mechanically, but the monitoring literature has not.

Practical recipe (what a lab could run tomorrow, at their scale, for a named capability): identify the precursor circuit; log its strength densely alongside a behavioral probe; manufacture blocked negatives by ablating the capability’s training signal; calibrate an anchored conformal interval on a seed fleet; freeze; monitor; and re-certify after any recipe change — expecting, per our tradeoff, that the FA-control features are the first casualties of shift.

8 Limitations

One capability (induction), small models (2-layer fleets; ≤ 410 m public suites), synthetic fleet language, one blind config shift (two axes bundled), $n=30/10/18$ per role. The fleet’s positive-vs-negative separation was clean partly because a bigram language makes the precursor repetition-specific; harder languages may reintroduce the trap our grokking rungs documented. R4’s calibration seeds were not blind (R5 is). The per-seed spread cleared its pre-registered bar by 0.01. Pythia event times are bounded by public checkpoint granularity. All pre-registration is repository-native (commit-stamped, publicly pushed), not third-party.

9 Reproducibility

Every number in this paper is a macro generated from scored artifacts (`paper4/gen_numbers.py`, byte-verified by `paper4/verify_regen.py`). The prereg chain is public: plan and kills before code (2cf62e3), forecaster freezes before test blocks, fleet spec before launch (6b05770), the blind gate frozen verbatim before its runs existed (377511b), verdicts after (f25aa45, dd4d5f5, 2ca7da4). Corpora: 466 grokking runs, 5 public-suite probe trajectories, 45+13 fleet runs with

dense circuit probes; training and scoring harnesses: `src/train_lm.py`, `src/probe_pythia.py`, `analysis/score_r*_p6.py`. Hardware: one RTX 3080.

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